

Simulation of high-power laser spectra using generative machine learning

Bc. Anna Kapitánová
Supervisor: Ing. Matěj Jech

Faculty of Information Technology
Czech Technical University in Prague

June 10, 2025

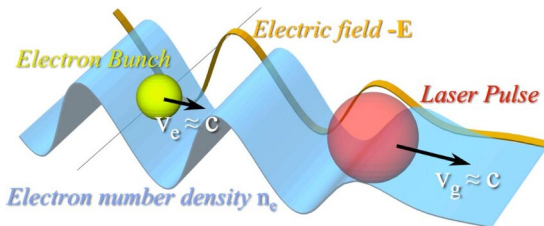


- Part of the **European research initiative Extreme Light Infrastructure ELI ERIC**
- Houses four **high-power lasers**, producing **ultra-short pulses**, which last only a few femtoseconds (10^{-15} seconds)
- Advantage of ultra-short pulses is their ability to produce **extremely high peak power** – reaching terawatts (TW), for the L4 laser, even petawatts (PW)



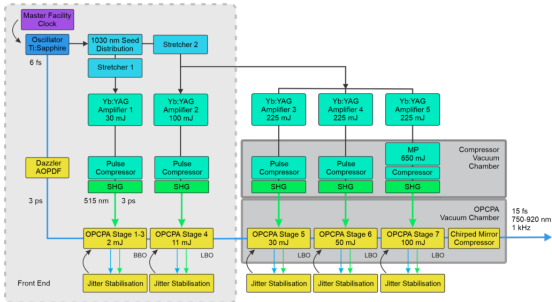
Laser Wakefield Acceleration (LWFA)

- Technique where a **laser pulse creates a wave in the plasma that rapidly accelerates electrons** to high energies over very short distances
- Applications in very high-energy radiotherapy, x-ray generation, ultra-fast radiobiology, radiochemistry, or studies on radiation effects in electronics



Laser L1 Allegra

- One of the lasers developed at ELI beamlines
- Produces high-energy, ultra-short laser pulses using a technique called Optical Parametric Chirped Pulse amplification (OPCPA)
- ALFA (Allegra Laser For Acceleration) is a laser-plasma electron accelerator



Problem Statement

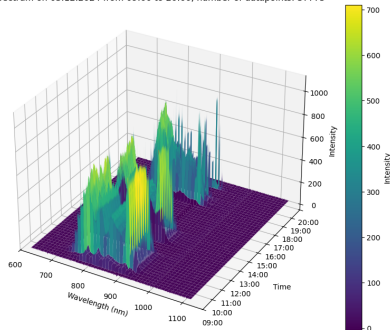
- **Prediction of Spectrum Intensities with Diffusion models**

- Simulation of spectral outcomes before running costly experiments
- Useful for filling in missing data blocks

- **Prediction of Ideal Parameters using Bayesian optimization**

- Define target values for the key spectral characteristics (the mean wavelength, the position of the maximum intensity, and the maximum intensity value)

Spectrum on 03.12.2024 from 09:00 to 20:00, number of datapoints: 37775



Missing spectral blocks

- Laser spectra output might be affected by many small, unmeasured factors – the goal is to generate multiple realistic outcomes, capturing uncertainty and variability
- Diffusion models are a class of generative models that learn to create data by gradually adding and removing noise
- Models used:
 - **Elucidated Diffusion Model (EDM)**
 - **Denoising Diffusion Implicit Model (DDIM)**
 - The best performing EDM model, selected based on validation metrics (MSE, Wasserstein Distance, Dynamic Time Warping)
 - Trained with a dropout rate of 0 %, and evaluated using 30 noise steps and a fixed CFG scale of 2

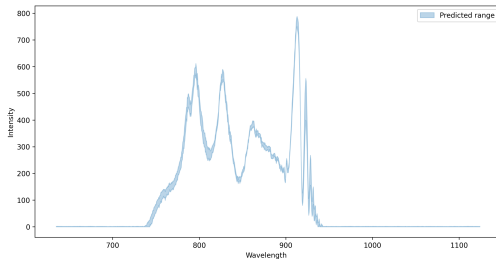
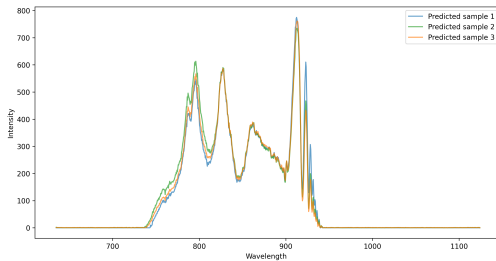
User Interface – Spectrum Intensities

L1-PL3-PZM52_Setpoint	-0.612	Usual range: (-3.00, 3.00)
L1-PL3-PZM53_Setpoint	0.1487	Usual range: (-3.00, 3.00)
L1-PL3-PZM56_Setpoint	-0.4105	Usual range: (-1.59, 0.63)
L1-PL3-PZM57_Setpoint	-1.2972	Usual range: (-3.58, -0.02)
L1-PL3-PZM58_Setpoint	-1	Usual range: (-3.00, 1.00)
L1-PL3-PZM59_Setpoint	-1	Usual range: (-3.00, 1.00)
L1-PL4-PZM66_Setpoint	-2.4977	Usual range: (-3.75, 3.11)
L1-PL4-PZM67_Setpoint	-0.4915	Usual range: (-3.75, 0.28)
L1-PL4-PZM68_Setpoint	-2.5347	Usual range: (-3.75, 3.75)
L1-PL4-PZM69_Setpoint	0.7122	Usual range: (-1.98, 3.75)
L1-INJ-PM70:VAL_CAL	31.08	Usual range: (-0.35, 41.89)
L1-OPA3-5_2-PM98:VAL_CAL	43.41	Usual range: (5.62, 66.25)
L1-INJ-PM70:VAL_CAL_diff	0	Usual range: (-23.72, 1.08)

n_samples

Sample

User Interface – Spectrum Intensities



User Interface – Parameters

✓ Select All

✗ Deselect All

Select parameters to optimize and set values for fixed parameters:

☐ L1-PL3-PZM52_Setpoint	-0.612	usual range: [-3.0, 3.0]
☒ L1-PL3-PZM53_Setpoint	0.1487	usual range: [-3.0, 3.0]
☐ L1-PL3-PZM56_Setpoint	-0.4105	usual range: [-1.591, 0.6349]
☒ L1-PL3-PZM57_Setpoint	-1.2972	usual range: [-3.5833, -0.0177]
☒ L1-PL3-PZM58_Setpoint	-1	usual range: [-3.0, 1.0]
☒ L1-PL3-PZM59_Setpoint	-1	usual range: [-3.0, 1.0]
☒ L1-PL4-PZM66_Setpoint	-2.4977	usual range: [-3.75, 3.1082]
☐ L1-PL4-PZM67_Setpoint	-0.4915	usual range: [-3.75, 0.2802]
☐ L1-PL4-PZM68_Setpoint	-2.5347	usual range: [-3.75, 3.75]
☐ L1-PL4-PZM69_Setpoint	0.7122	usual range: [-1.9848, 3.75]
☐ L1-INJ-PM70:VAL_CAL	31.08	usual range: [-0.35, 41.89]
☐ L1-OPA3-5_2-PM98:VAL_CAL	43.41	usual range: [5.62, 66.25]
☐ L1-INJ-PM70:VAL_CAL_diff	0	usual range: [-33.72, 1.08]

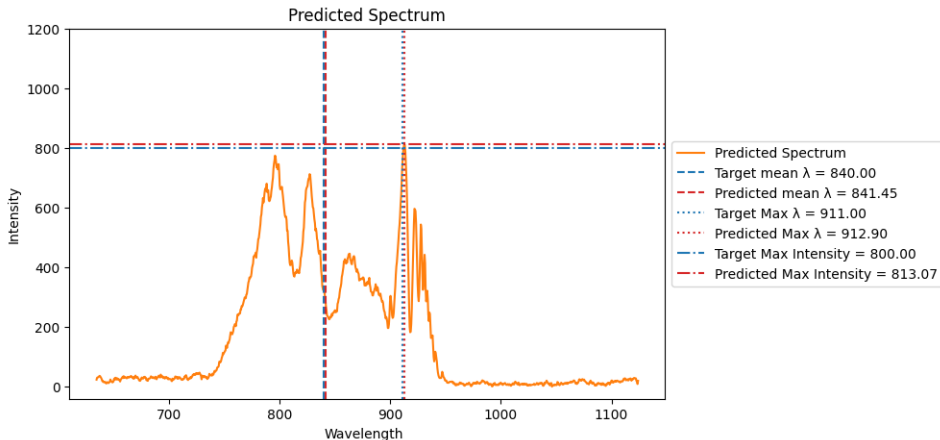
Target spectrum properties:

Target Mean λ	840	usual range: [830, 850]
Target Max λ	911	usual range: [794, 913]
Target Max Intensity	800	usual range: [600, 820]

n_calls: 10

▶ Run Optimization

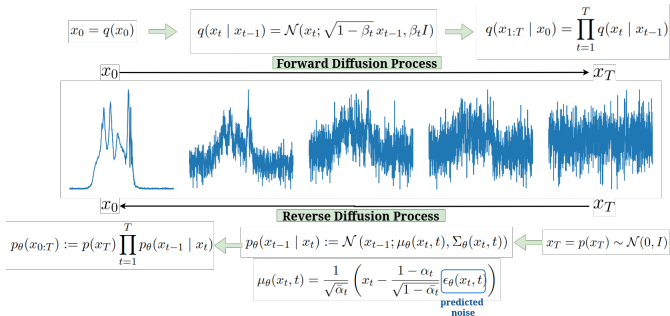
User Interface – Parameters



Thank You for Your Attention

Diffusion Probabilistic Models

- Diffusion models are a class of generative models that learn to create data by gradually adding and removing noise



Denoising Diffusion Implicit Model (DDIM) [2]

- A deterministic variant derived from *Denoising Diffusion Probabilistic Model (DDPM)* [1], designed to speed up sampling.
- Direct transitions between distant time steps – significantly reduces the number of steps required while preserving output quality.
- The DDIM sampling update rule is:

$$x_{t-1} = \sqrt{\alpha_{t-1}} \left(\frac{x_t - \sqrt{1 - \alpha_t} \epsilon_\theta(x_t)}{\sqrt{\alpha_t}} \right) + \sqrt{1 - \alpha_{t-1} - \sigma_t^2} \cdot \epsilon_\theta(x_t) + \sigma_t \epsilon_t$$

where ϵ_t is Gaussian noise independent of x_t .

- The flexibility in choosing σ_t allows deterministic ($\sigma_t = 0$) or stochastic sampling at timestep t :

$$\sigma_t = \sqrt{\frac{1 - \alpha_{t-1}}{1 - \alpha_t}} \cdot \sqrt{\frac{1 - \alpha_t}{\alpha_{t-1}}}$$

Elucidated Diffusion Model (EDM) [4]

- Builds on ideas from *Score-Based Generative Modeling through Stochastic Differential Equations (SDEs)* [3]
- Main idea:
 - Forward diffusion processes are **solutions of SDEs**
 - For reverse diffusion process, represented by the reverse SDE, it is necessary to estimate the **score function** $\nabla_x \log p_t(x)$, which is the gradient of the log-probability with respect to the data
 - The score function is **approximated by a neural network** $s_\theta(x, t)$ at different noise levels (score-based generative modeling)

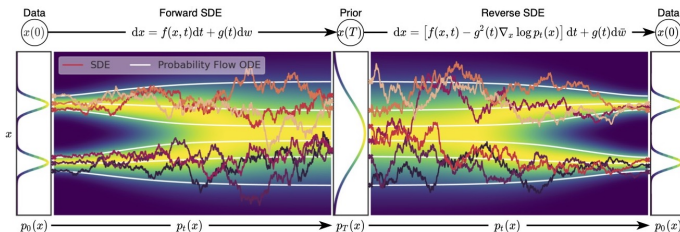


Figure: Score-based generative modeling through SDEs (Source: [3]).

- **Mean Square Error (MSE)**, gives a point-by-point comparison

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

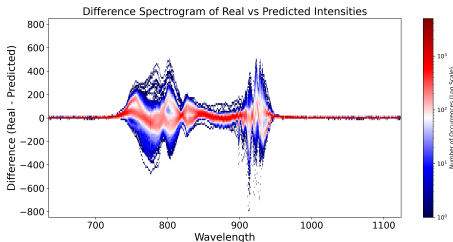
- **Dynamic Time Warping (DTW)**, compares two sequences X and Y by allowing position shifts [5]

$$\text{DTW}(X, Y) = \min_{\pi} \left(\sum_{(i,j) \in \pi} d(x_i, y_j) \right)$$

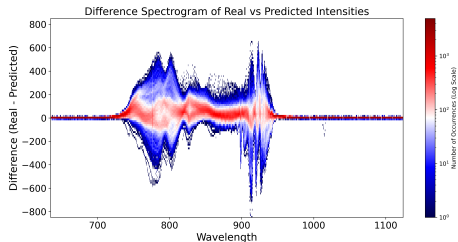
- **Wasserstein Distance (Earth Mover's Distance)**, measures the "cost" of transforming distribution P into Q by moving probability mass [6]

$$W(P, Q) = \inf_{\gamma \in \Gamma(P, Q)} \int_{\mathbb{R} \times \mathbb{R}} |x - y| d\gamma(x, y)$$

EDM and DDIM Comparison



EDM



DDIM

Model	MSE		EM		DTW	
	Mean	Median	Mean	Median	Mean	Median
EDM	2265.93	2051.32	16.67	5.61	9658.38	9581.03
DDIM	4606.29	2864.54	24.41	7.64	14291.10	12371.86

Table: Comparison of evaluation metrics between EDM and DDIM.

Best-Performing Model

- The best performing EDM model, selected based on validation metrics
- Trained with a dropout rate of 0 %, and evaluated using 30 noise steps and a fixed CFG scale of 2

Metric	Mean	Median	Min	Max
MSE	2612.67	2502.20	46.40	30428.93
DTW	9504.94	9403.86	3055.29	41077.11
EM	20.90	6.55	0.18	146.14

Table: Evaluation metrics of the best EDM model on the test dataset.

Bayesian Optimization Framework

- **Goal:** Tune laser system parameters to match a target spectrum.
- **Model:** Conditional EDM diffusion model generates spectra from parameter vectors.
- **Optimization Target:** Minimize differences in:
 - Mean wavelength: $\lambda_{\text{mean}} = \frac{\sum_{\lambda} I(\lambda) \cdot \lambda}{\sum_{\lambda} I(\lambda)}$
 - Peak wavelength λ_{max} (where intensity is maximal)
 - Maximum intensity I_{max}

- **Loss Function:**

$$\text{Loss} = |\lambda_{\text{mean}}^{\text{gen}} - \lambda_{\text{mean}}^{\text{target}}| + |\lambda_{\text{max}}^{\text{gen}} - \lambda_{\text{max}}^{\text{target}}| + |I_{\text{max}}^{\text{gen}} - I_{\text{max}}^{\text{target}}|$$

- **Search Space:** Subset of 13 parameters chosen by user.
- **Method:** `gp_minimize` with `gp_hedge` strategy in `scikit-optimize`.
- **gp_hedge:** dynamically chooses between three commonly used acquisition functions: Lower Confidence Bound, Expected Improvement, and Probability of Improvement.

References I

-  HO, Jonathan; JAIN, Ajay; ABBEEL, Pieter. Denoising diffusion probabilistic models. *Advances in neural information processing systems*. 2020, vol. 33, pp. 6840–6851.
-  SONG, Jiaming; MENG, Chenlin; ERMON, Stefano. Denoising diffusion implicit models. *arXiv preprint arXiv:2010.02502*. 2020.
-  SONG, Yang; SOHL-DICKSTEIN, Jascha; KINGMA, Diederik P; KUMAR, Abhishek; ERMON, Stefano; POOLE, Ben. Score-based generative modeling through stochastic differential equations. *arXiv preprint arXiv:2011.13456*. 2020.
-  KARRAS, Tero; AITTALA, Miika; AILA, Timo; LAINE, Samuli. Elucidating the design space of diffusion-based generative models. *Advances in neural information processing systems*. 2022, vol. 35, pp. 26565–26577.

References II



RABINER, Lawrence; JUANG, Biing-Hwang. *Fundamentals of speech recognition*. Prentice-Hall, Inc., 1993.



VILLANI, Cédric et al. *Optimal transport: old and new*. Vol. 338. Springer, 2008.